Advanced VLSI Design - Lab 5

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1 Schematic and Layout of a NAND2 Circuit

A NANDN configuration is specified by having PMOS transistors in parallel and NMOS transistors in series. The schematic of the NAND2 configuration can be seen in *Figure 1.* To find the switching point of a NAND2 configuration, we can use the following equation:

$$V_{sp} = \frac{\sqrt{\frac{\beta_n}{4\beta_p}} \cdot V_{THN} + (V_{DD} - V_{THP})}{1 + \sqrt{\frac{\beta_n}{4\beta_p}}}$$

Where $k_{NAND2} = \frac{\beta_n}{4\beta_p}$, with higher order configurations written as $k_{NANDN} = \frac{\beta_n}{N^2\beta_p}$.

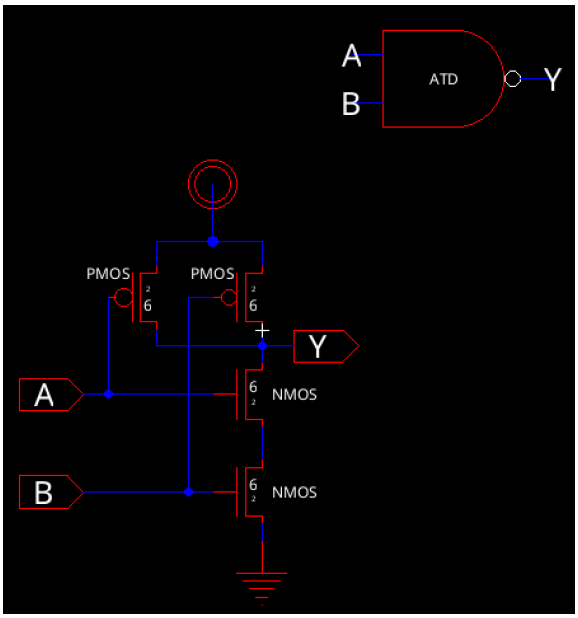


Figure 1: NAND2 Schematic with Corresponding Icon View

In standard applications, PMOS transistors are typically made larger than NMOS transistors due to electron mobility differences. However, this schematic presents both the PMOS and NMOS as being identical sizes. The transconductance can then be represented by $\beta_n = 3\beta_p$, further simplifying our equation into:

$$V_{sp} = \frac{\sqrt{\frac{3}{4}} \cdot V_{THN} + (V_{DD} - V_{THP})}{1 + \sqrt{\frac{3}{4}}}$$

Thus, as N increases, our ratio k_{NAND2} decreases, and the overall switching point voltage of the NAND2 configuration can be estimated to be above the ideal switching point.

The switching point transient simulation was tested using the following LTSpice code:

```
vdd vdd 0 dc 5
vin d_in 0 pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n
.include *C5_models.txt
```

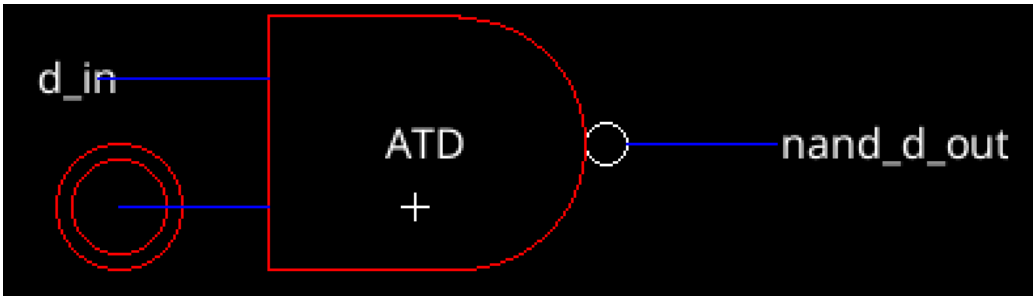


Figure 2: NAND2 Transient Simulation Schematic

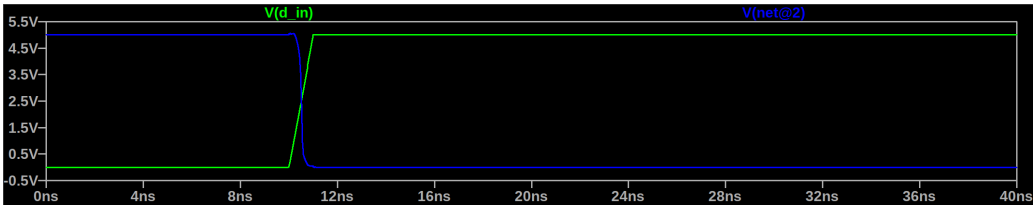


Figure 3: NAND2 Transient LTspice Simulation Results

A NAND2 truth table was simulated on LTspice by injecting a pulse wave into a and b. The following code was used to simulate this truth table:

```
vdd vdd 0 dc 5
va 0 pulse(0v 5v 0n 1n 1n 10n 20n)
```

```
vb pulse(0v 5v 5n 1n 1n 10n 20n)
.tran 0 40n
.include *C5_models.txt
```

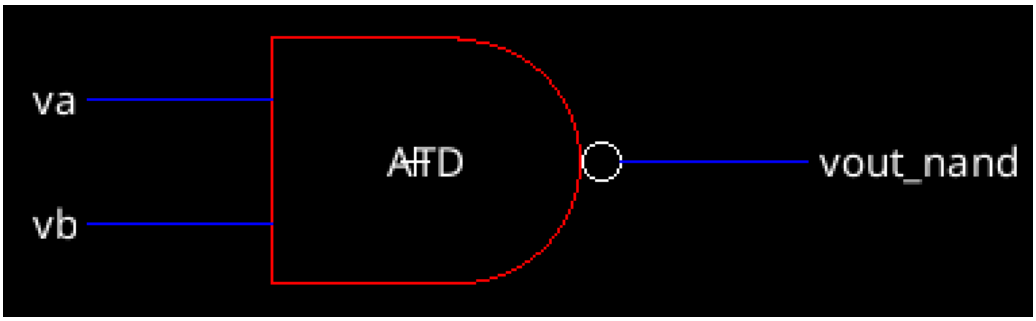


Figure 4: NAND2 Truth Table Simulation Schematic

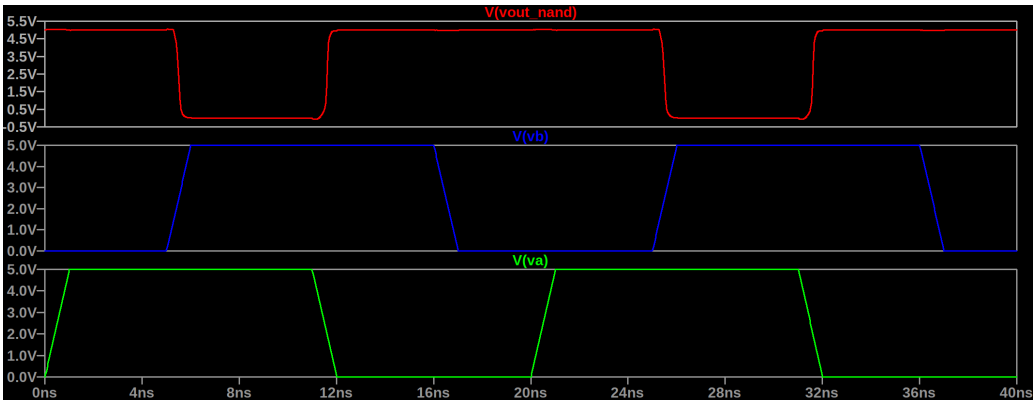


Figure 5: NAND2 Truth Table LTspice Simulation Results

1.1 Layout of a NAND2 Gate

To create the layout for the NAND2 circuit, two nMOS and two pMOS components were added to the layout window followed by their corresponding Acts and Wells. For the PMOS transistors, both drains were tied to the output while both sources were tied to VDD. For the NMOS transistors, the drain of NMOS1 was tied the ouput while its source was tied to the drain of NMOS2. The source of NMOS2 was tied to ground.

Metal-1 connections were used to tie sources and drains while polysilicon-1 was used to tie the gates to their respective input.

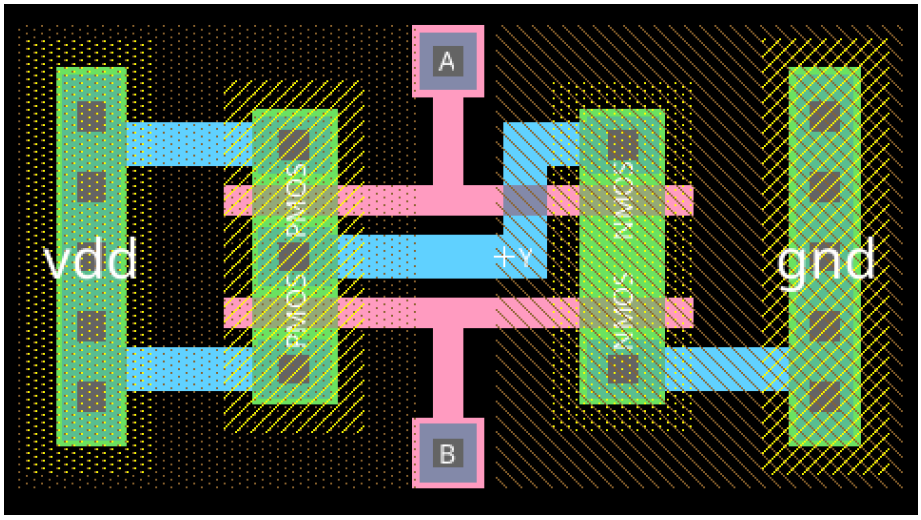


Figure 6: NAND2 Layout

2 Schematic and Layout of a NOT Circuit

The schematic for the NOT gate was created by placing a PMOS and NMOS in series with both of their gates tied to the input and both drains ties to the output. The source of the PMOS was tied to VDD and source of NMOS tied to ground.

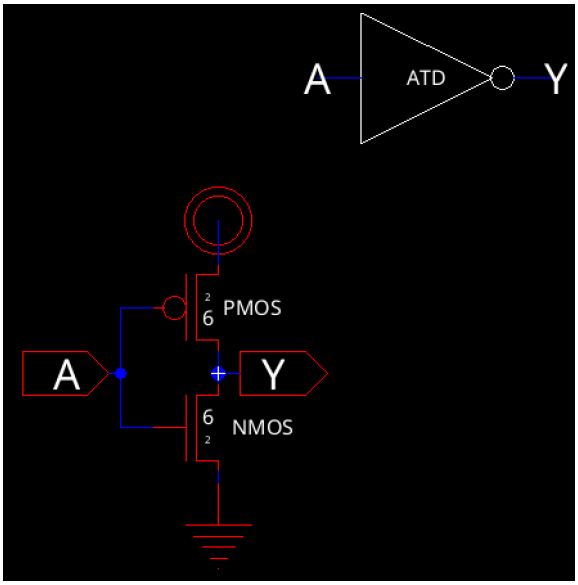


Figure 7: NOT Gate Schematic with Corresponding Icon View

A switching point simulation was run using the identical LTspice code used to simulated the NAND2 switching point:

```
vdd vdd 0 dc 5
vin d_in pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n
.include *C5_models.txt
```

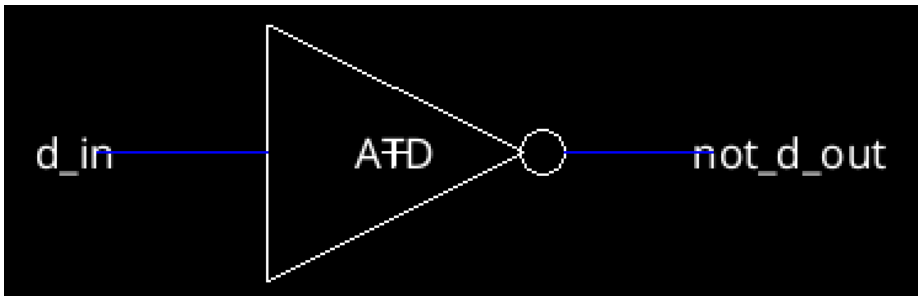


Figure 8: NOT Transient Simulation Schematic

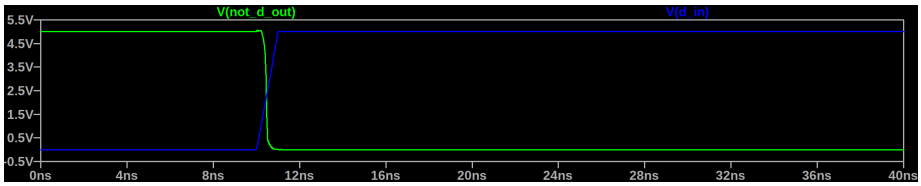


Figure 9: NOT Gate LTspice Transient Simulation Results

2.1 Layout of NOT Circuit

The layout for a NOT circuit was created by placing identical sized pMOS and nMOS components in the layout window. Acts and Wells were placed with according to their respective transistors. Drains and sources were connected using Metal-1 wire, while the input was tied to the gates via polysilicon-1 connections.

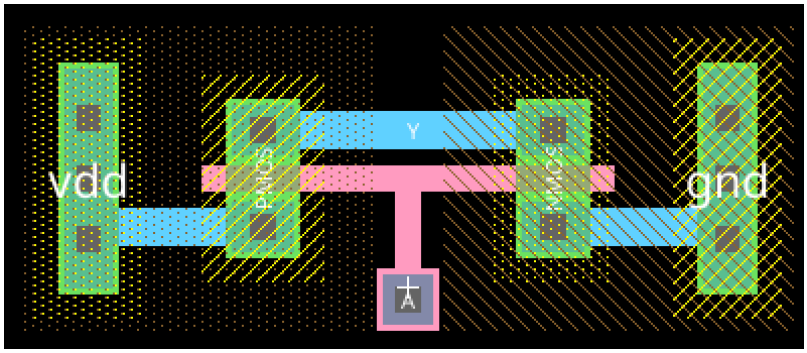


Figure 10: NOT Gate Layout

3 Schematic and Layout of an Exclusive OR Gate

The exclusive OR gate schematic consisted of parallel and series configurations of both PMOS and NMOS transistors with inverted and non-inverted inputs coming from standard NOT gates.

The circuit can be derived from a boolean algebra expression using DeMorgan's Theorem:

$$y = A\overline{B} + B\overline{A}$$
$$\overline{y} = \overline{A\overline{B} + B\overline{A}}$$
$$\overline{y} = (\overline{A} + B)(\overline{B} + A)$$
$$\overline{\overline{y}} = y = \overline{(\overline{A} + B)(\overline{B} + A)}$$

Where in the PMOS configuration, OR $(+)$ is a series connection and AND $(\cdot)$ is a parallel connection. In the NMOS configuration, the OR $(+)$ and AND $(\cdot)$ are swapped.

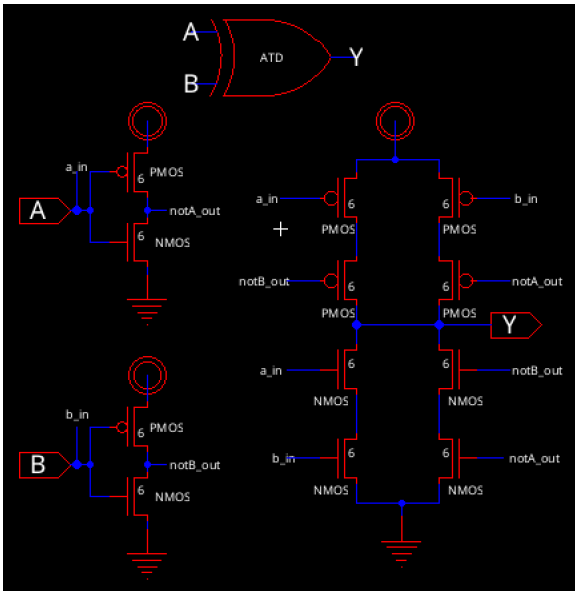


Figure 11: XOR2 Schematic with Corresponding Icon View

A transient simulation was run in LTspice using the following code:

```
vdd vdd 0 dc 5
vin d_in 0 pulse(0v 5v 10n 1n 1n 40n 40n)
.tran 0 40n
.include *C5_models.txt
```

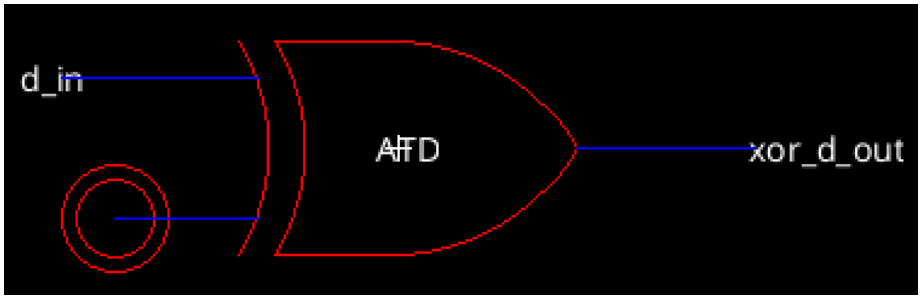


Figure 12: XOR2 Transient Simulation Schematic

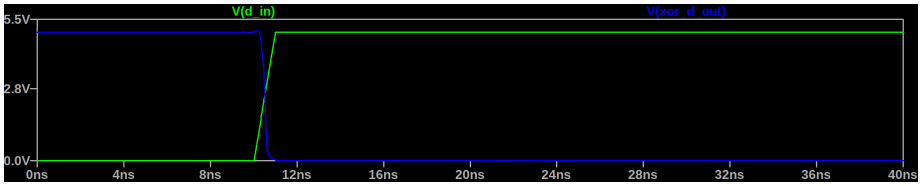


Figure 13: XOR2 Transient Simulation LTspice Results

A truth table simulation was also run by injecting a pulse wave into A and B:

```
vdd vdd 0 dc 5
va va 0 pulse(0v 5v 0n 1n 1n 10n 20n)
vb vb 0 pulse(0v 5v 5n 1n 1n 10n 20n)
.tran 0 40n
.include *C5_models.txt
```

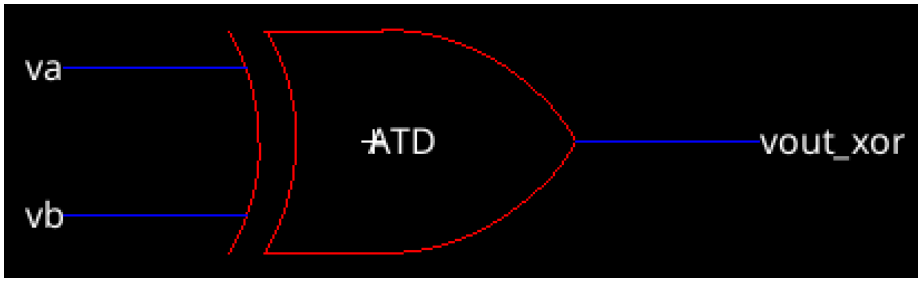


Figure 14: XOR2 Truth Table Simulation Schematic

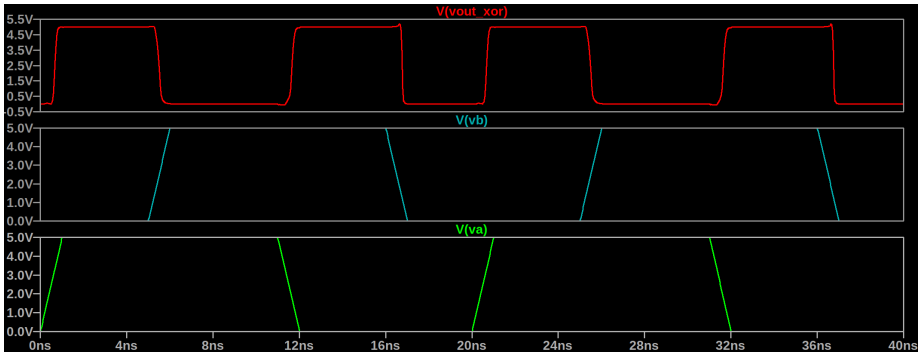


Figure 15: XOR2 Truth Table LTspice Simulation Results

3.1 Layout of an Exclusive OR Gate

Creating the layout for the exclusive OR gate required placing two NOT gates (discussed in section. . .) to establish the inputs. The exclusive OR utilized 4 nMOS, 5 nActs, 4 pMOS, 3 pActs, 1

nWell, and 1 pWell. Source and drain connections were made using Metal-1 wires, while the Metal-2 lines connected the drains to sources of the NMOS configuration. The drains of two NMOS and two PMOS were tied via Metal-1 wires tied to a Metal-1 to Metal-2 Pin. All gates were tied via polysilicon-1.

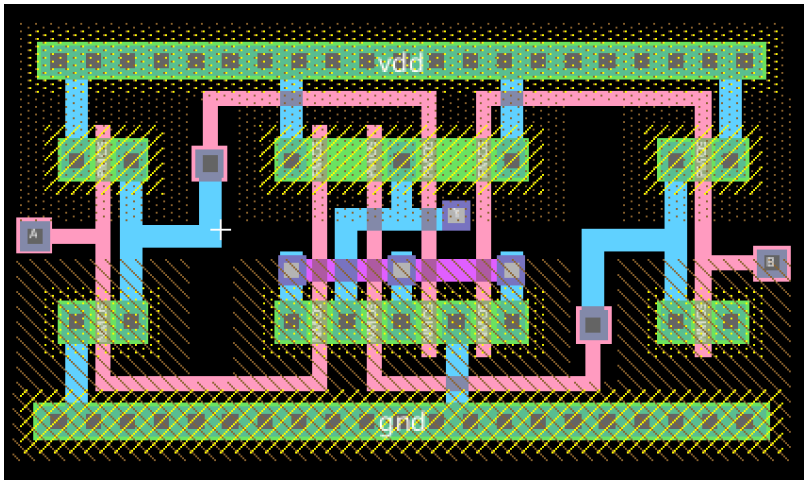


Figure 16: XOR2 Layout

4 Schematic and Layout of a Full Adder Circuit

To create the full adder circuit, two exclusive ORs and three NAND gates were used. The schematic takes three inputs: A, B, C_{in} and outputs value S and C_{out}. A full adder simulation was run in LTspice using the following code:

```
vdd vdd 0 dc 5
va a 0 pulse(0v 5v 0n 1n 1n 10n 20n)
vb b 0 pulse(0v 5v 5n 1n 1n 10n 20n)
vcin cin 0 pulse(0v 5v 2n 1n 1n 10n 20n)
.tran 0 60n
.include *C5_models.txt
```

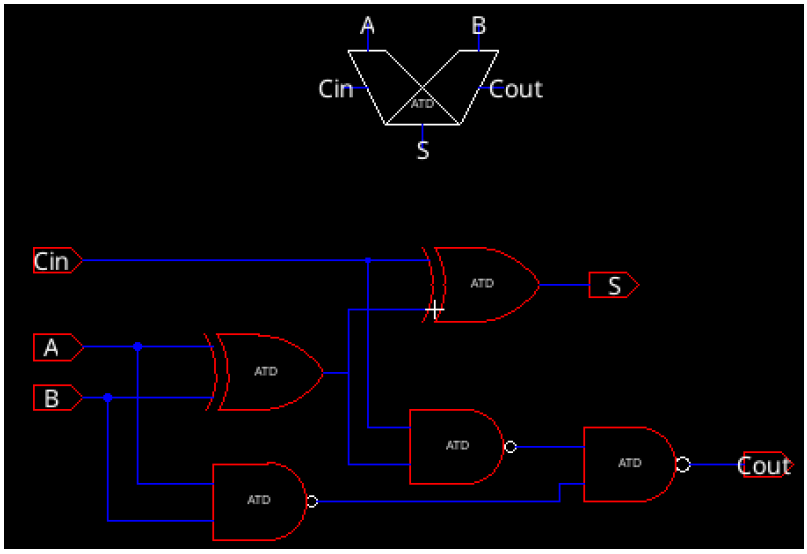


Figure 17: Full Adder Schematic with Corresponding Icon View

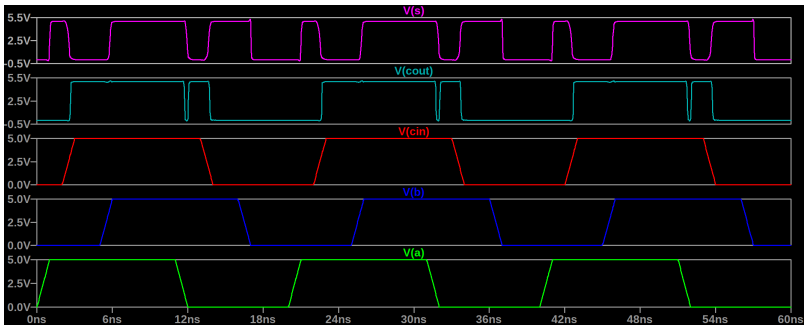


Figure 18: Full Adder LTspice Simulation Results

4.1 Layout of Full Adder Circuit

As stated in the schematic, this layout utilized two exclusive or and three NAND layouts. Sources and Drains were tied using Metal-1 connections and Metal-2 connections where necessary to prevent wire overlap. Figure 19 demonstrates the full layout.

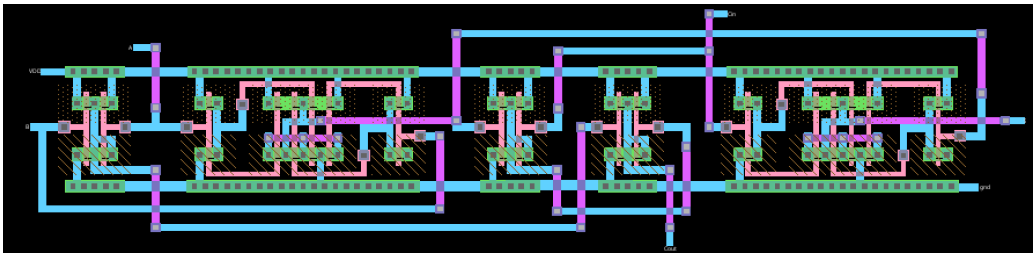


Figure 19: Full Adder Layout

The greatest challenge when producing this layout was ensuring no errors occurred during wiring. Two regions in specific were flagged for wiring errors and were note easily resolved. Visually, the two regions do no pose concern for errors; however MOSIS rules flagged it as follows. The primary regions of error are noted in figure 20 and 21.

```
Notch (layer 'Metal-1'):  LESS (BY 2) THAN 3; (Y axis) [rule '7.2 Mosis, SUBM']
```

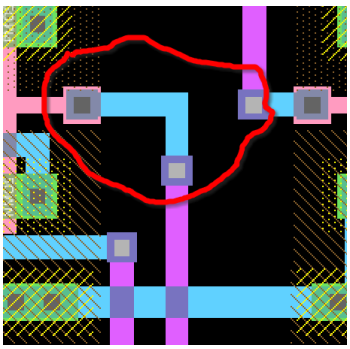


Figure 20: Caption for first image

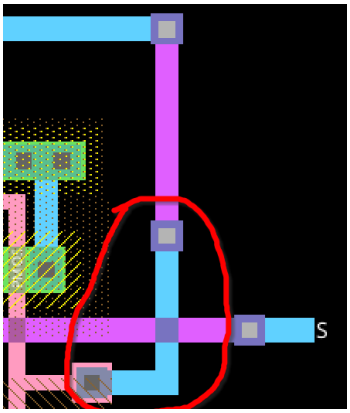


Figure 21: Caption for second image

5 Documentation